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CHASSIS, CARRIAGE AND LOCK MODULE

Abstract

A chassis includes a casing and a carriage. The casing has an engaging portion. The carriage includes a housing and a lock module. The lock module includes a bracket, a damping member, a fixing structure and a lock member. The bracket has an accommodating portion and an engaging recess. The damping member is disposed in the accommodating portion. The damping member has a hole portion. The fixing structure is disposed in the hole portion. The lock module is fixed on the housing via the fixing structure. The lock member is movably disposed on the bracket. When the engaging portion is located in the engaging recess and the lock member is located in the engaging recess, the lock member restrains the engaging portion in the engaging recess to lock the carriage on the casing.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The disclosure relates to a chassis, a carriage and a lock module and, more particularly, to a lock module having vibration damping function and capable of being assembled or disassembled conveniently, and a chassis and a carriage equipped with the lock module.

2. Description of the Prior Art

[0002] The current structural design of servers often fixes a mother board or other circuit boards (e.g. printed circuit board assembly (PCBA)) on a tray, and then assembles the tray to a casing of a chassis. In order to facilitate disassembly and assembly, the tray usually adopts a toolless design, such that the tray and the casing of the chassis cannot be tightly fixed. During shock and vibration test, vibrations will be transmitted to the tray through the casing of the chassis, thereby causing solder crack or other damages on the circuit board.

SUMMARY OF THE INVENTION

[0003] The disclosure provides a lock module having vibration damping function and capable of being assembled or disassembled conveniently, and a chassis and a carriage equipped with the lock module, so as to solve the aforesaid problems.

[0004] According to an embodiment of the disclosure, a chassis comprises a casing and a carriage. The casing has an engaging portion. The carriage comprises a housing and a lock module. The lock module comprises a bracket, a damping member, a fixing structure and a lock member. The bracket has an accommodating portion and an engaging recess. The damping member is disposed in the accommodating portion. The damping member has a hole portion. The fixing structure is disposed in the hole portion. The lock module is fixed on the housing via the fixing structure. The lock member is movably disposed on the bracket. When the engaging portion is located in the engaging recess and the lock member is located in the engaging recess, the lock member restrains the engaging portion in the engaging recess to lock the carriage on the casing.

[0005] According to another embodiment of the disclosure, a carriage comprises a housing and a lock module. The lock module comprises a bracket, a damping member, a fixing structure and a lock member. The bracket has an accommodating portion and an engaging recess. The damping member is disposed in the accommodating portion. The damping member has a hole portion. The fixing structure is disposed in the hole portion. The lock module is fixed on the housing via the fixing structure. The lock member is movably disposed on the bracket.

[0006] According to another embodiment of the disclosure, a lock module comprises a bracket, a damping member, a fixing structure and a lock member. The bracket has an accommodating portion and an engaging recess. The damping member is disposed in the accommodating portion. The damping member has a hole portion. The fixing structure is disposed in the hole portion. The lock member is movably disposed on the bracket.

[0007] As mentioned in the above, the disclosure disposes the damping member between the bracket and the fixing structure to isolate the bracket from the fixing structure by the damping member. The damping member is capable of effectively absorbing vibrations transmitted from the casing of the chassis and the bracket, so as to reduce the vibration energy transmitted from the casing of the chassis to the carriage. Accordingly, the disclosure can prevent vibration from causing damage to a circuit board fixed on the carriage. Furthermore, the lock module restrains the engaging portion in the engaging recess by the lock member, so as to lock the carriage on the casing. As long as the lock member moves away from the engaging recess, the engaging member is

able to disengage from the engaging recess, so as to detach the carriage from the casing. Accordingly, a user can conveniently and rapidly disassemble or assemble the carriage. [0008] These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is an exploded view illustrating a chassis according to an embodiment of the disclosure.

[0010] FIG. 2 is a partial perspective view illustrating the chassis shown in FIG. 1.

[0011] FIG. 3 is a perspective view illustrating a lock module shown in FIG. 2 from another viewing angle.

[0012] FIG. 4 is an exploded view illustrating the lock module shown in FIG. 3.

[0013] FIG. 5 is a perspective view illustrating an operating member shown in FIG. 3 moving to a first position.

[0014] FIG. 6 is a perspective view illustrating a carriage shown in FIG. 1 being put into a casing.

[0015] FIG. 7 is a partial perspective view illustrating the chassis shown in FIG. 6.

[0016] FIG. 8 is a perspective view illustrating the operating member shown in FIG. 6 moving to the first position.

[0017] FIG. 9 is a partial perspective view illustrating the chassis shown in FIG. 8.

[0018] FIG. 10 is a front view illustrating a lock module according to another embodiment of the disclosure.

[0019] FIG. 11 is a front view illustrating the engaging portion pushing the lock member to rotate and moving into the engaging recess.

[0020] FIG. 12 is a front view illustrating the engaging portion pushing the lock member to rotate and moving out of the engaging recess.

[0021] FIG. 13 is a perspective bottom view illustrating the lock module shown in FIG. 10.

DETAILED DESCRIPTION

[0022] Referring to FIGS. 1 to 9, FIG. 1 is an exploded view illustrating a chassis 1 according to an embodiment of the disclosure, FIG. 2 is a partial perspective view illustrating the chassis 1 shown in FIG. 1, FIG. 3 is a perspective view illustrating a lock module 122 shown in FIG. 2 from another viewing angle, FIG. 4 is an exploded view illustrating the lock module 122 shown in FIG. 3, FIG. 5 is a perspective view illustrating an operating member 1228 shown in FIG. 3 moving to a first position, FIG. 6 is a perspective view illustrating a carriage 12 shown in FIG. 1 being put into a casing 10, FIG. 7 is a partial perspective view illustrating the chassis 1 shown in FIG. 6, FIG. 8 is a perspective view illustrating the operating member 1228 shown in FIG. 6 moving to the first position, and FIG. 9 is a partial perspective view illustrating the chassis 1 shown in FIG. 8.

[0023] As shown in FIGS. 1 and 2, the chassis 1 comprises a casing 10 and a carriage 12. It should be noted that the casing 10 shown in FIG. 1 is only a part of a complete casing of the chassis 1. The chassis 1 may be a chassis of a server or other devices according to practical applications. The casing has an engaging portion 100. In this embodiment, the casing 10 may have four engaging portions 100 respectively disposed at two sides of the casing 10. The carriage 12 comprises a housing 120 and a lock module 122. In this embodiment, the carriage 12 may comprise four lock modules 122 respectively disposed at two sides of the housing 120, wherein the positions of the four lock modules 122 respectively correspond to the positions of the four engaging portions 100. It should be noted that the number and position of the lock modules 122 and the engaging portions 100 may be determined according to practical applications, so the disclosure is not limited to the

embodiment shown in the figure. In this embodiment, two sides of the housing **120** may have four recesses **1200** for accommodating the four lock modules **122**. In this embodiment, the carriage **12** may be a tray and the lock module **122** may be disposed on a side wall of the tray, but the disclosure is not so limited.

[0024] As shown in FIGS. 2 to 5, the lock module **122** comprises a bracket **1220**, a damping member **1222**, a fixing structure **1224**, a lock member **1226**, an operating member **1228**, an elastic member **1230** and a cover **1232**. The bracket **1220** has an accommodating portion **12200** and an engaging recess **12202**. The damping member **1222** is disposed in the accommodating portion **12200** of the bracket **1220** and the damping member **1222** has a hole portion **12220**. The fixing structure **1224** is disposed in the hole portion **12220** of the damping member **1222**, such that the damping member **1222** isolates the bracket **1220** from the fixing structure **1224**. The bracket **1220** and the fixing structure **1224** may be made of aluminum alloy or other materials. The damping member **1222** may be made of rubber or other materials capable of absorbing vibration. For example, when the damping member **1222** is made of rubber, the bracket **1220**, the damping member **1222** and the fixing structure **1224** may be jointed during rubber molding process, but the disclosure is not so limited.

[0025] The lock module **122** is fixed on the housing **120** via the fixing structure **1224**. As shown in FIG. 2, a fixing member **124** (e.g. screw) may be inserted into the fixing structure **1224** and fixed to the housing **120**, so as to fix the lock module **122** on the housing **120**. After the lock module **122** is fixed on the housing **120**, the bracket **1220** is not in contact with the housing **120**.

[0026] The lock member **1226** is movably disposed on the bracket **1220** and corresponds to the engaging recess **12202**. In this embodiment, the bracket **1220** may have a sliding groove **12204** and the lock member **1226** is movably disposed in the sliding groove **12204**. The cover **1232** is disposed on the bracket **1220** and covers the sliding groove **12204**, so as to prevent the lock member **1226** and the elastic member **1230** from coming off the bracket **1220**. The elastic member **1230** is also disposed in the sliding groove **12204**, and two ends of the elastic member **1230** abut against the lock member **1226** and the cover **1232**. In this embodiment, the bracket **1220** may have two sliding grooves **12204** located at two sides of the engaging recess **12202**. Furthermore, the lock module **122** may comprise two lock members **1226**, two elastic members **1230** and two covers **1232**, wherein the two lock members **1226** and the two elastic members **1230** are respectively disposed in the two sliding grooves **12204**, and the two covers **1232** respectively cover the two sliding grooves **12204**.

[0027] In this embodiment, the lock member **1226** may have an inclined surface **12260**. The lock member **1226** may move with respect to the bracket **1220** between the positions shown in FIGS. 3 and 5. When the lock member **1226** is located at the position shown in FIG. 3, the lock member **1226** moves away from the engaging recess **12202** and the inclined surface **12260** protrudes from the bracket **1220**. When the lock member **1226** is located at the position shown in FIG. 5, the lock member **1226** is located in the engaging recess **12202** and the inclined surface **12260** retracts into the bracket **1220**.

[0028] The operating member **1228** is movably disposed on the bracket **1220**. In this embodiment, the operating member **1228** may have two arm portions **12280** and a handle portion **12282**, wherein the two arm portions **12280** extend from two sides of the handle portion **12282**, and the operating member **1228** is U-shaped. Furthermore, the handle portion **12282** may protrude with respect to the two arm portions **12280** to form an operating space **12284** between the two arm portions **12280**. The operating member **1228** is able to move with respect to the bracket **1220** between a second position shown in FIG. 3 and a first position shown in FIG. 5. In this embodiment, a moving direction of the lock member **1226** is perpendicular to a moving direction of the operating member **1228**, but the disclosure is not so limited.

[0029] In this embodiment, each of the two arm portions **12280** may have a first restraining portion **12286** and a second restraining portion **12288**, and the bracket **1220** may have two third restraining

portions **12206** and two fourth restraining portions **12208**. As shown in FIG. 5, when the operating member **1228** is located at the first position, the first restraining portion **12286** can cooperate with the third restraining portion **12206** to restrain the operating member **1228** at the first position. A user may extend a finger into the operating space **12284** and push the handle portion **12282** toward a direction of an arrow **A1** to disengage the first restraining portion **12286** from the third restraining portion **12206**. Then, the user may pull the handle portion **12282** upward to move the operating member **1228** to the second position shown in FIG. 3. As shown in FIG. 3, when the operating member **1228** is located at the second position, the second restraining portion **12288** can cooperate with the fourth restraining portion **12208** to restrain the operating member **1228** at the second position. The user may tilt the handle portion **12282** toward the direction of the arrow **A1** to disengage the second restraining portion **12288** from the fourth restraining portion **12208**. Then, the user may detach the operating member **1228** from the bracket **1220**. Similarly, the user may inclinedly insert the arm portion **12280** of the operating member **1228** into a gap between the third restraining portion **12206** and the fourth restraining portion **12208** to assemble the operating member **1228** to the bracket **1220**.

[0030] Before assembling the carriage **12** to the casing **10**, the operating member **1228** needs to be kept at the second position, as shown in FIGS. 1 and 2. At this time, the elastic member **1230** (as shown in FIG. 4) keeps the lock member **1226** at the position shown in FIG. 2. Thus, when the operating member **1228** is located at the second position, the inclined surface **12260** of the lock member **1226** will protrude from the bracket **1220**. Then, a user may align the engaging recess **12202** of the bracket **1220** with the engaging portion **100** of the casing **10** to put the carriage **12** into the casing **10**, as shown in FIGS. 6 and 7. At this time, the engaging portion **100** is located in the engaging recess **12202**. Then, the user may press the operating member **1228** downward, such that the operating member **1228** moves from the second position (as shown in FIGS. 6 and 7) to the first position (as shown in FIGS. 8 and 9). When the operating member **1228** moves from the second position to the first position, the operating member **1228** pushes the inclined surface **12260**, such that the lock member **1226** moves into the engaging recess **12202**. Thus, as shown in FIGS. 8 and 9, when the engaging portion **100** is located in the engaging recess **12202** and the lock member **1226** is located in the engaging recess **12202**, the lock member **1226** restrains the engaging portion **100** in the engaging recess **12202** to lock the carriage **12** on the casing **10**. Furthermore, when the operating member **1228** is located at the first position, the operating member **1228** restrains the lock member **1226** from moving, such that the lock member **1226** is located in the engaging recess **12202**. Accordingly, the carriage **12** can be stably locked on the casing **10**.

[0031] The lock module **122** is fixed on the housing **120** via the fixing structure **1224**, engages with the engaging portion **100** of the casing **10** through the engaging recess **12202** of the bracket **1220**, and isolates the bracket **1220** from the fixing structure **1224** through the damping member **1222**. The damping member **1222** is capable of effectively absorbing vibrations transmitted from the casing **10** and the bracket **1220**, so as to reduce the vibration energy transmitted from the casing **10** to the carriage **12**. Accordingly, the disclosure can prevent vibration from causing damage to a circuit board fixed on the carriage **12**.

[0032] When the operating member **1228** is located at the first position (as shown in FIG. 9), the lock member **1226** compresses the elastic member **1230** (as shown in FIG. 4). When the user wants to detach the carriage **12** from the casing **10**, the user may push the handle portion **12282** of the operating member **1228** toward the direction of the arrow (as shown in FIG. 5) to disengage the first restraining portion **12286** from the third restraining portion **12206**. Then, the user may pull the handle portion **12282** upward to move the operating member **1228** to the second position shown in FIG. 7. When the operating member **1228** moves from the first position to the second position, the elastic member **1230** (as shown in FIG. 4) drives the lock member **1226** to return, such that the lock member **1226** moves away from the engaging recess **12202**. Thus, when the operating member **1228** is located at the second position, the operating member **1228** disengages from the lock

member **1226**, such that the lock member **1226** is able to move away from the engaging recess **12202** and the engaging portion **100** is able to disengage from the engaging recess **12202**. Then, the user may pull the handle portion **12282** of the operating member **1228** upward to detach the carriage **12** from the casing **10**.

[0033] Referring to FIGS. **10** to **13**, FIG. **10** is a front view illustrating a lock module **122'** according to another embodiment of the disclosure, FIG. **11** is a front view illustrating the engaging portion **100** pushing the lock member **1226** to rotate and moving into the engaging recess **12202**, FIG. **12** is a front view illustrating the engaging portion **100** pushing the lock member **1226** to rotate and moving out of the engaging recess **12202**, and FIG. **13** is a perspective bottom view illustrating the lock module **122'** shown in FIG. **10**.

[0034] The main difference between the lock module **122'** and the aforesaid lock module **122** is that the lock member **1226** of the lock module **122'** is pivotally connected to the bracket **1220**, such that the engaging portion **100** is able to push the lock member **1226** to rotate and move into or out of the engaging recess **12202**, as shown in FIGS. **10** to **12**. The lock module **122** shown in FIG. **1** may be replaced by the lock module **122'**, so as to utilize the lock module **122'** to lock the carriage **12** on the casing **10**. In this embodiment, a double torsion spring (not shown) may be disposed on the lock member **1226**. As shown in FIG. **11**, when the engaging recess **12202** of the bracket **1220** is aligned with the engaging portion **100** and moves downward, the engaging portion **100** will push the lock member **1226** to rotate and move into the engaging recess **12202**. At this time, the lock member **1226** moves away from the engaging recess **12202**. Then, the double torsion spring drives the lock member **1226** to return to a horizontal position shown in FIG. **10**. At this time, the lock member **1226** is located in the engaging recess **12202** to restrain the engaging portion **100** in the engaging recess **12202**. Furthermore, as shown in FIG. **12**, when the bracket **1220** moves upward, the engaging portion **100** will push the lock member **1226** to rotate and move out of the engaging recess **12202**. At this time, the lock member **1226** also moves away from the engaging recess **12202**. Then, the double torsion spring drives the lock member **1226** to return to the horizontal position shown in FIG. **10**.

[0035] In this embodiment, the operating member **1228** may have an avoidance recess **12290**. As shown in FIG. **13**, when the operating member **1228** is located at the first position, an end **12262** of the lock member **1226** protruding from the bracket **1220** is accommodated in the avoidance recess **12290**. At this time, the engaging portion **100** shown in FIG. **10** cannot push down the lock member **1226** to rotate. Thus, when the operating member **1228** is located at the first position, the operating member **1228** restrains the lock member **1226** from rotating, such that the lock member **1226** is located in the engaging recess **12202**. In other words, the operating member **1228** needs to move from the first position (as shown in FIG. **10**) to the second position (as shown in FIG. **11**) before the engaging portion **100** can push the lock member **1226** to rotate and move out of the engaging recess **12202**.

[0036] As mentioned in the above, the disclosure disposes the damping member between the bracket and the fixing structure to isolate the bracket from the fixing structure by the damping member. The damping member is capable of effectively absorbing vibrations transmitted from the casing of the chassis and the bracket, so as to reduce the vibration energy transmitted from the casing of the chassis to the carriage. Accordingly, the disclosure can prevent vibration from causing damage to a circuit board fixed on the carriage. Furthermore, the lock module restrains the engaging portion in the engaging recess by the lock member, so as to lock the carriage on the casing. As long as the lock member moves away from the engaging recess, the engaging member is able to disengage from the engaging recess, so as to detach the carriage from the casing. Accordingly, a user can conveniently and rapidly disassemble or assemble the carriage.

[0037] Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly,

the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A chassis comprising: a casing having an engaging portion; and a carriage comprising a housing and a lock module, the lock module comprising: a bracket having an accommodating portion and an engaging recess; a damping member disposed in the accommodating portion, the damping member having a hole portion; a fixing structure disposed in the hole portion, the lock module being fixed on the housing via the fixing structure; and a lock member movably disposed on the bracket; wherein, when the engaging portion is located in the engaging recess and the lock member is located in the engaging recess, the lock member restrains the engaging portion in the engaging recess to lock the carriage on the casing.
2. The chassis of claim 1, wherein the lock module further comprises an operating member movably disposed on the bracket; wherein, when the operating member is located at a first position, the operating member restrains the lock member from moving, such that the lock member is located in the engaging recess; wherein, when the operating member is located at a second position, the operating member disengages from the lock member, such that the lock member is able to move away from the engaging recess and the engaging portion is able to disengage from the engaging recess.
3. The chassis of claim 2, wherein the lock member has an inclined surface; wherein, when the operating member is located at the second position, the inclined surface protrudes from the bracket; wherein, when the operating member moves from the second position to the first position, the operating member pushes the inclined surface, such that the lock member moves into the engaging recess.
4. The chassis of claim 2, wherein the operating member has two arm portions and a handle portion, the two arm portions extend from two sides of the handle portion, and the operating member is U-shaped.
5. The chassis of claim 4, wherein each of the two arm portions has a first restraining portion and a second restraining portion, the bracket has two third restraining portions and two fourth restraining portions, the first restraining portion cooperates with the third restraining portion to restrain the operating member at the first position, and the second restraining portion cooperates with the fourth restraining portion to restrain the operating member at the second position.
6. The chassis of claim 2, wherein the bracket has a sliding groove, the lock member is movably disposed in the sliding groove, and a moving direction of the lock member is perpendicular to a moving direction of the operating member.
7. The chassis of claim 6, wherein the lock module further comprises an elastic member disposed in the sliding groove; wherein, when the operating member is located at the first position, the lock member compresses the elastic member; wherein, when the operating member moves from the first position to the second position, the elastic member drives the lock member to return, such that the lock member moves away from the engaging recess.
8. The chassis of claim 2, wherein the lock member is pivotally connected to the bracket, the engaging portion is able to push the lock member to move into the engaging recess, and the operating member has an avoidance recess; wherein, when the operating member is located at the first position, an end of the lock member protruding from the bracket is accommodated in the avoidance recess.
9. The chassis of claim 1, wherein the carriage is a tray and the lock module is disposed on a side wall of the tray.
10. A carriage comprising: a housing; and a lock module comprising: a bracket having an accommodating portion and an engaging recess; a damping member disposed in the

accommodating portion, the damping member having a hole portion; a fixing structure disposed in the hole portion, the lock module being fixed on the housing via the fixing structure; and a lock member movably disposed on the bracket.

11. The carriage of claim 10, wherein the lock module further comprises an operating member movably disposed on the bracket; wherein, when the operating member is located at a first position, the operating member restrains the lock member from moving, such that the lock member is located in the engaging recess; wherein, when the operating member is located at a second position, the operating member disengages from the lock member, such that the lock member is able to move away from the engaging recess.

12. The carriage of claim 11, wherein the lock member has an inclined surface; wherein, when the operating member is located at the second position, the inclined surface protrudes from the bracket; wherein, when the operating member moves from the second position to the first position, the operating member pushes the inclined surface, such that the lock member moves into the engaging recess.

13. The carriage of claim 11, wherein the operating member has two arm portions and a handle portion, the two arm portions extend from two sides of the handle portion, and the operating member is U-shaped.

14. The carriage of claim 13, wherein each of the two arm portions has a first restraining portion and a second restraining portion, the bracket has two third restraining portions and two fourth restraining portions, the first restraining portion cooperates with the third restraining portion to restrain the operating member at the first position, and the second restraining portion cooperates with the fourth restraining portion to restrain the operating member at the second position.

15. The carriage of claim 11, wherein the bracket has a sliding groove, the lock member is movably disposed in the sliding groove, and a moving direction of the lock member is perpendicular to a moving direction of the operating member.

16. The carriage of claim 11, wherein the lock member is pivotally connected to the bracket, the lock member is able to move into the engaging recess, and the operating member has an avoidance recess; wherein, when the operating member is located at the first position, an end of the lock member protruding from the bracket is accommodated in the avoidance recess.

17. A lock module comprising: a bracket having an accommodating portion and an engaging recess; a damping member disposed in the accommodating portion, the damping member having a hole portion; a fixing structure disposed in the hole portion; and a lock member movably disposed on the bracket.

18. The lock module of claim 17, wherein the lock module further comprises an operating member movably disposed on the bracket, and the lock member has an inclined surface; wherein, when the operating member is located at a first position, the operating member restrains the lock member from moving, such that the lock member is located in the engaging recess; wherein, when the operating member is located at a second position, the inclined surface protrudes from the bracket and the operating member disengages from the lock member, such that the lock member is able to move away from the engaging recess; wherein, when the operating member moves from the second position to the first position, the operating member pushes the inclined surface, such that the lock member moves into the engaging recess.

19. The lock module of claim 18, wherein the operating member has two arm portions and a handle portion, the two arm portions extend from two sides of the handle portion, and the operating member is U-shaped.

20. The lock module of claim 19, wherein each of the two arm portions has a first restraining portion and a second restraining portion, the bracket has two third restraining portions and two fourth restraining portions, the first restraining portion cooperates with the third restraining portion to restrain the operating member at the first position, and the second restraining portion cooperates with the fourth restraining portion to restrain the operating member at the second position.

